

Chapter 4

Structured Analysis

Structured Analysis

- Structured Analysis is a development method that allows the analyst to understand the system and its activities in a logical way.
- It is a systematic approach, which uses graphical tools that analyze and refine the objectives of an existing system and develop a new system specification which can be easily understandable by user.

It has following attributes –

- It is **graphic** which specifies the presentation of application.
- It **divides the processes** so that it gives a clear picture of system flow.
- It is **logical rather than physical** i.e., the elements of system do not depend on vendor or hardware.
- It is an approach that works from **high-level overviews to lower-level details**.

Structured Analysis Tools

- During Structured Analysis, various tools and techniques are used for system development.
- They are :
 - Data Flow Diagrams
 - Data Dictionary
 - Decision Trees
 - Decision Tables

Data Flow Diagram

- DFD is graphical representation/model of flow of data in an information system.
- It is capable of representing incoming data flow, outgoing data flow and stored data.
- Data Flow Diagrams are either Logical or Physical.
- **Logical DFD** - It concentrates on the system process, and flow of data in the system.
 - ✓ eg, in a Banking software system, how data is moved between different entities.
- **Physical DFD** - shows how the data flow is actually implemented in the system. It is more specific and close to the implementation.

DFD Components

- DFD can represent Source, destination, storage and flow of data using the following set of components -



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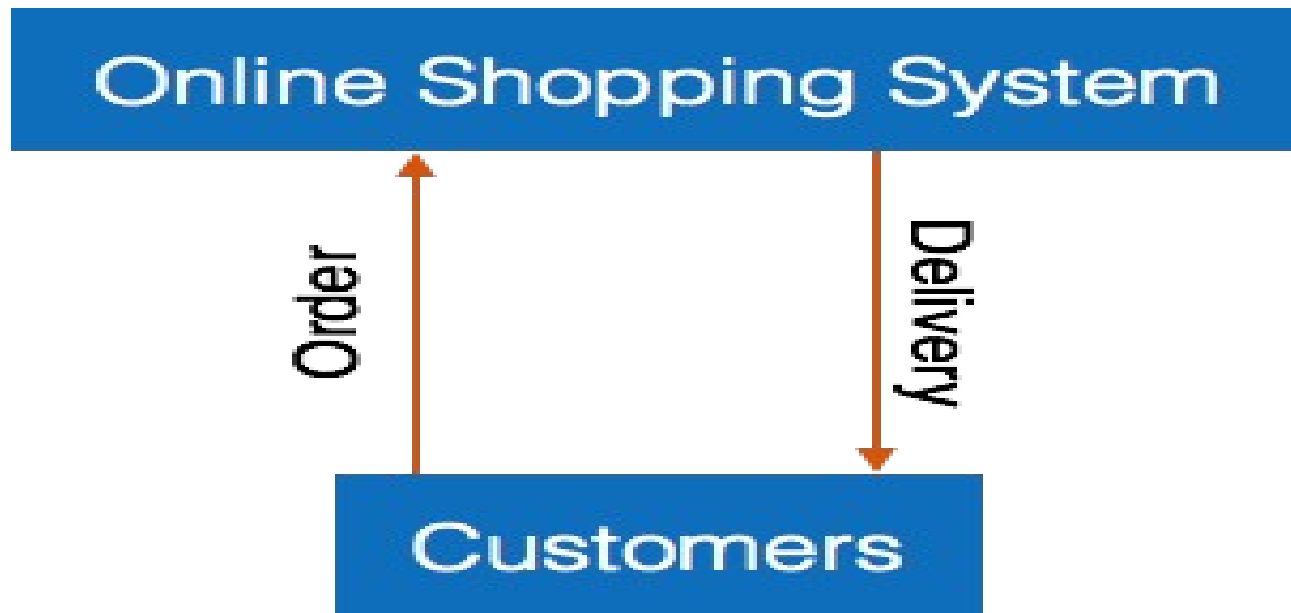
- **Entities** - Entities are **source and destination** of information data. Entities are represented by a **rectangles** with their respective names.
- **Process** - **Activities and action** taken on the data are represented by **Circle or Round-edged** rectangles.

Data Storage - A **location where data is saved to or retrieved from** , such as a data base

- **Data Flow** - **Movement of data & is shown by pointed arrows.**
Data movement is shown from the base of arrow as its **source** towards head of the arrow as **destination**.

Levels of DFD

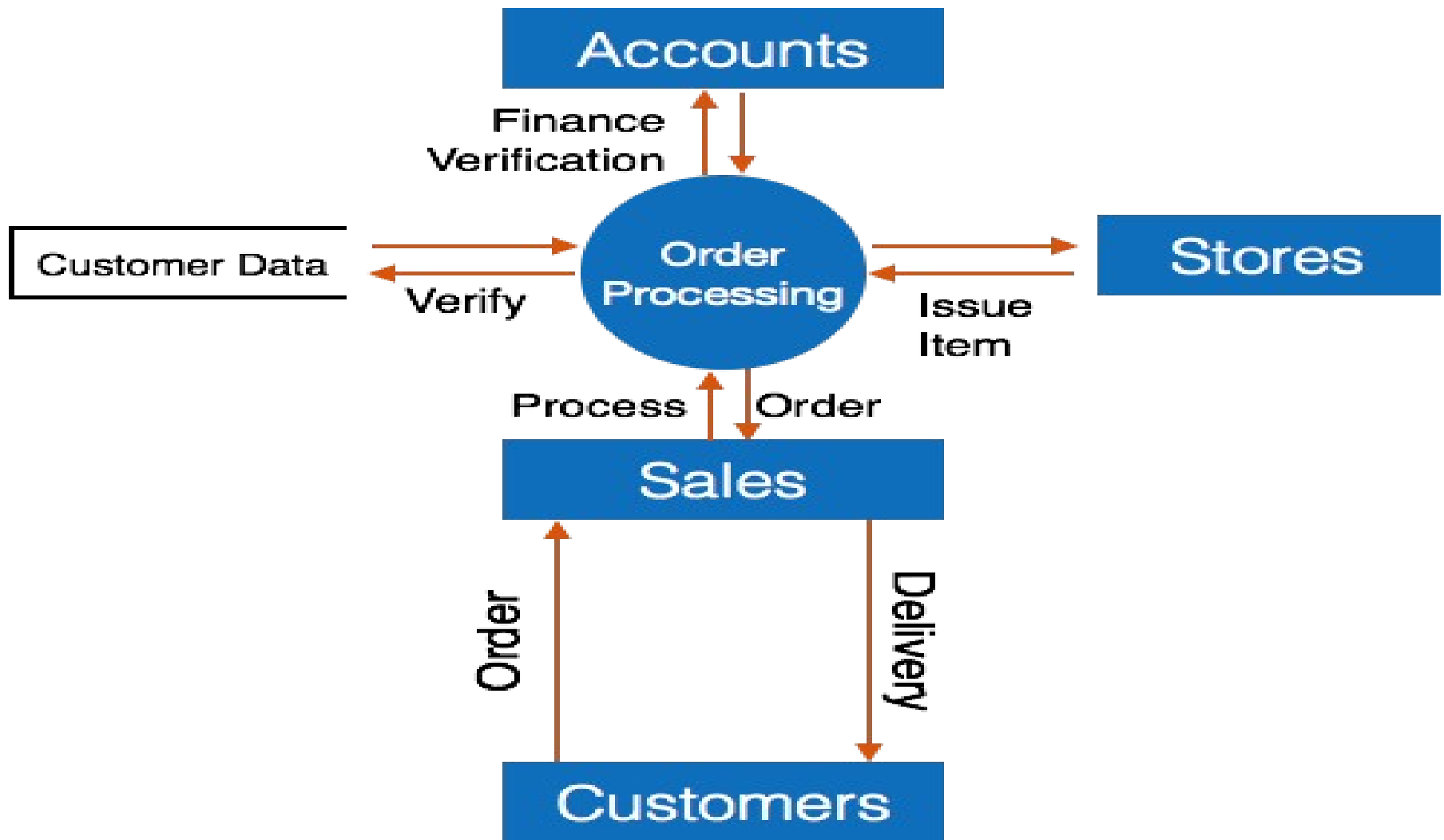
- **Level 0** - it is the **highest abstraction level** which depicts the **entire information system as one diagram** hiding all the underlying details.
they are also known **as context level DFDs**.



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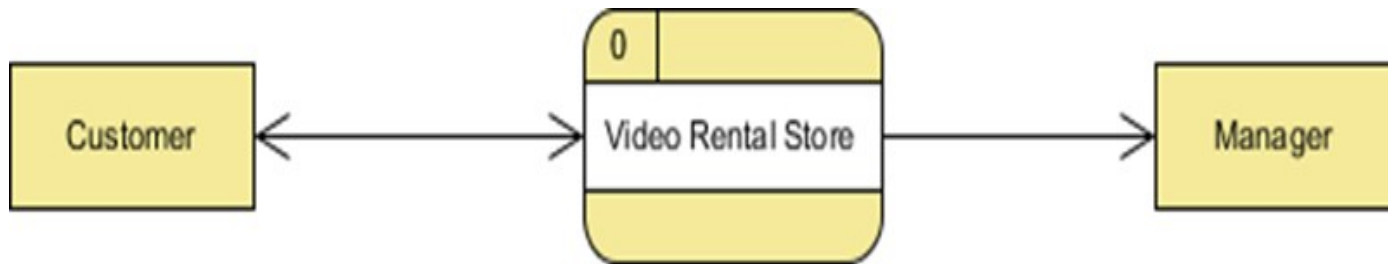
- **Level 1** - The Level 0 DFD is broken down into more specific, Level 1 DFD.
- Level 1 DFD represents basic modules in the system and flow of data among different modules.
- It also mentions basic processes and sources of information.

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Example: Video Rental System

- Context diagram :

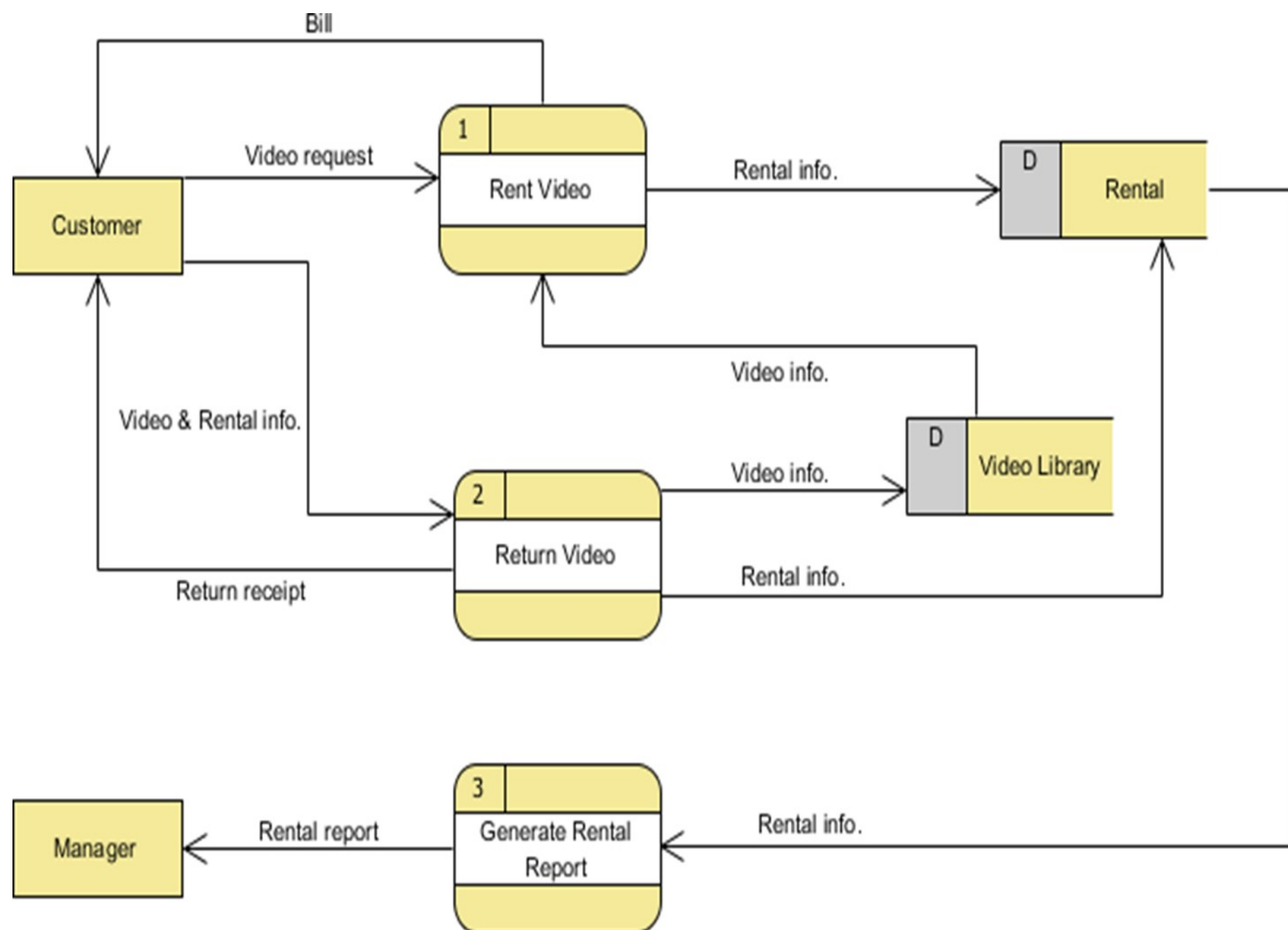


- ❑ a process (shape) = the system to model, in this case, the "Video Rental Store".
- ❑ External entities = the participants who will interact with the system. In this example, Customer and Manager.
- ❑ Data flow connector = the existence of information exchange between customer and the system.

Level 1 DFD

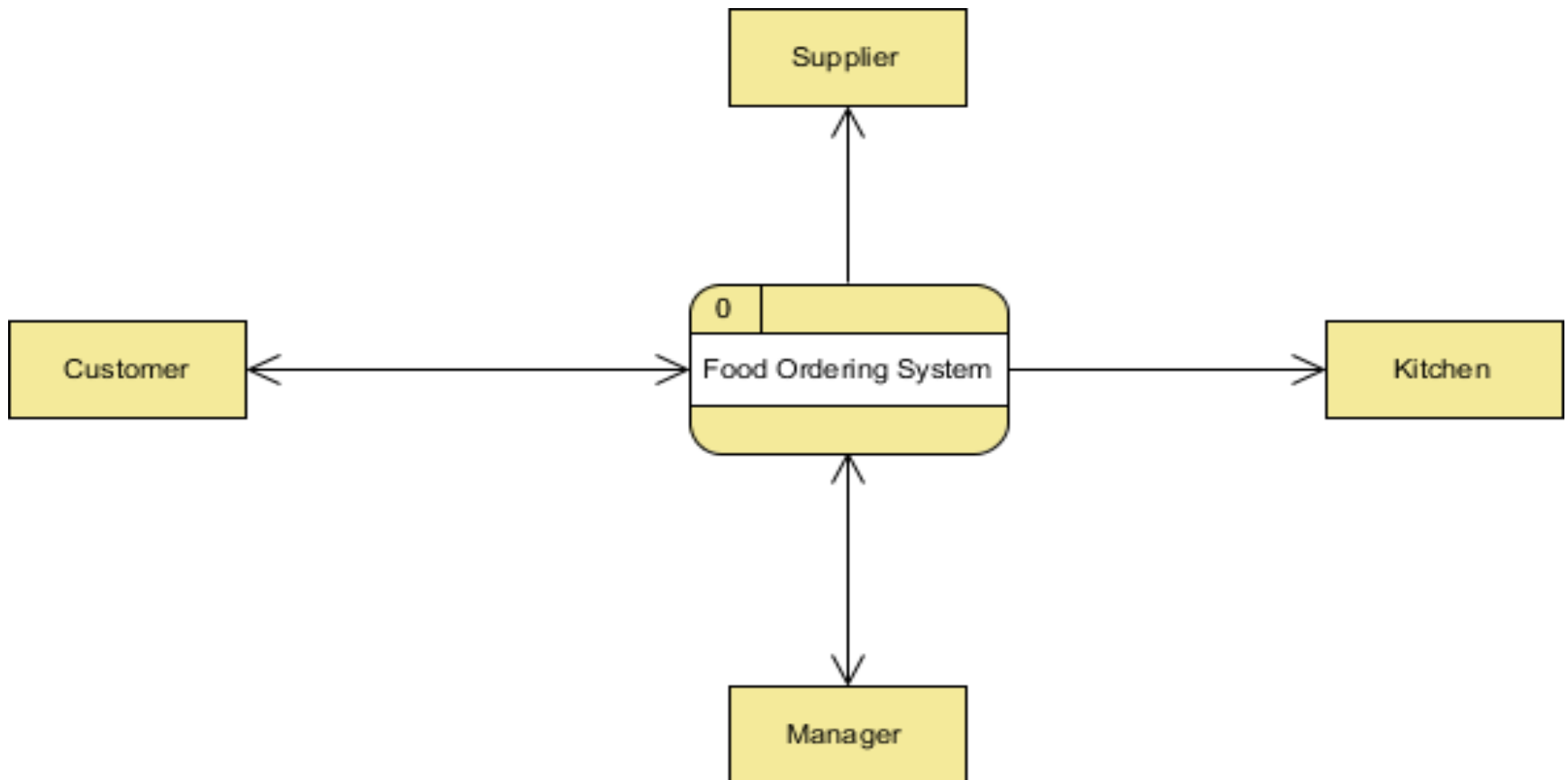
which is the decomposition (i.e. break down) of the video rental system that is shown in the context DFD.

- ✓ *Customer* makes a *Video request* to the *Rent Video* process.
 - ✓ The *Rent Video* process also receives *Video info.* from the *Video Library* data store. As a result, the process produces a *Bill* to the *Customer*, and stores the *Rental info.* into the *Rental* data store.
 - ✓ A *Customer* can *Return Video* by providing *Video & Rental info.*
 - ✓ The process stores the *Video info.* into the *Video Library* data store and *Rental info.* into the *Rental* data store. As a result, *Return receipt* is delivered to the *Customer*.
- “Return Video process receives Video & Rental info. and produces Video info., Rental info., and Return receipt, with no order specified.”*
- ✓ Finally, a *Manager* can receive *Rental report* from the *Generate Rental Report* process and the information involved is provided by the *Rental* data store.



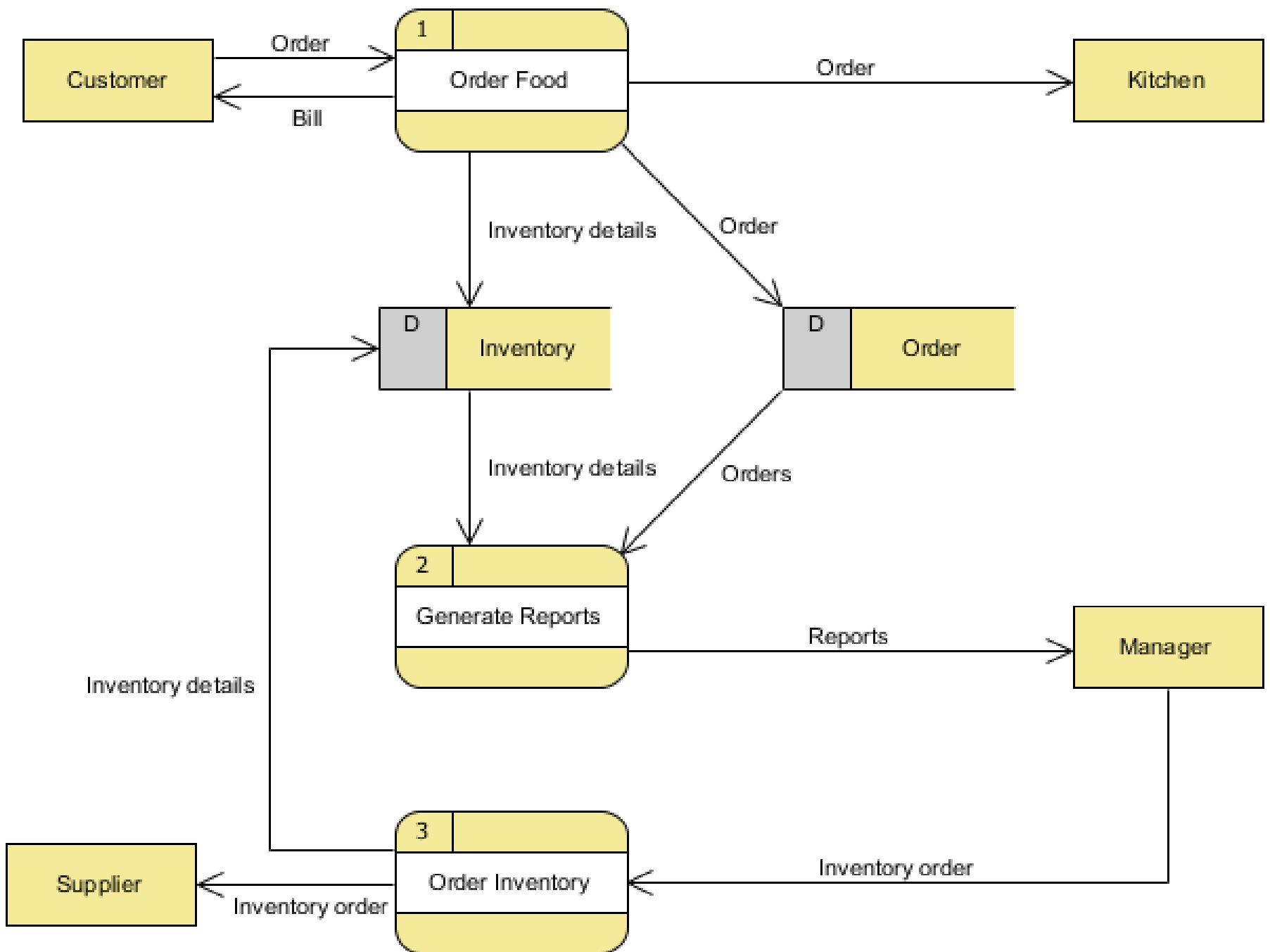
Example: Food ordering System

- Context DFD:



Level 1 DFD

- a *Customer* can place an *Order*. The *Order Food* process receives the *Order*, forwards it to the *Kitchen*, store it in the *Order* data store, and store the updated *Inventory details* in the *Inventory* data store. The process also delivers a *Bill* to the *Customer*.
- The *Manager* can receive *Reports* through the *Generate Reports* process, which takes *Inventory details* and *Orders* as input from the *Inventory* and *Order* data store respectively.
- The *Manager* can also initiate the *Order Inventory* process by providing *Inventory order*. The process forwards the *Inventory order* to the *Supplier* and stores the updated *Inventory details* in the *Inventory* data store.



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- **Level 2** - At this level, DFD shows how data flows inside the modules mentioned in Level 1.
- Higher level DFDs can be transformed into more specific lower level DFDs with deeper level of understanding unless the desired level of specification is achieved.

Data Dictionary

- Data dictionary is the centralized collection of information about data. It stores meaning and origin of data, its relationship with other data, data format for usage etc.
- It has solid definitions of all names in order to facilitate user and software designers.
- It is often referenced as meta-data (data about data) repository.
- It is created along with DFD model of software program and is expected to be updated whenever DFD is changed or updated.

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- The data is referenced using data dictionary while designing and implementing software.
- Data dictionary **removes** any chances of **ambiguity**.
- It helps keeping work of programmers and designers **synchronized** while using same object reference everywhere in the program.
- It also **provides a way of documentation for the complete database system** in one place.
- **Validation of DFD** is carried out using data dictionary.

Contents

- Data dictionary should contain information about the following
 - Data Flow
 - Data Structure
 - Data Elements
 - Data Stores

• Data Dictionary Contents

– Data Flow

- Each data flow should be defined with descriptive information and its composite structure or elements.
- It Includes the following information
 - » ID - identification number, Label, the text that should appear on the diagram.
 - A general description of the data flow.
 - The source of the data flow
 - » This could be an external **entity**,
 - » a **process**, or a data flow coming from a **data store**.
 - The destination of the data flow

Example

Name	Customer Order
Description	Contains customer order information and is used to update the customer master and item files and to produce an order record.
Source	Customer External Entity
Destination	Process 1, Add Customer Order
Type	Screen
Data Structure	Order Information
Volume/Time	10/hour
Comments	An order record contains information for one customer order. The order may be received by mail, fax, or by telephone.

Data Dictionary Contents

Data Structures

Data structures are a group of smaller structures and elements.

An algebraic notation is used to represent the data structure.

=	Composed of
{ }	Repetition
()	Optional
+	And
[/]	Or

Example

Address = House No + (Street / Area) + City + State

Course ID = Course Number + Course Name + Course Level + Course Grades

Data Dictionary Contents

Data Elements

- Data elements consist of **Name and descriptions of Data and Control Items, Internal or External data stores** etc. with the following details:
 - ✓ Primary Name
 - ✓ Secondary Name(Alias)
 - ✓ Use-case (How and where to use)
 - ✓ Content Description (Notation etc.)
 - ✓ Supplementary Information (preset values, constraints etc.)

Example

Name	Customer Number
Alias	Client Number
Alias	Receivable Account Number
Description	Uniquely identifies a customer that has made any business transaction within the last five years.
Length	6
Input Format	9(6)
Output Format	9(6)
Default Value	
Continuous/Discrete	Continuous
Type	Numeric
Base or Derived	Derived
Upper Limit	<999999
Lower Limit	>18
Discrete	Value/Meaning
Comments	The customer number must pass a modulus-11 check-digit test.

Data Dictionary Contents

Data Store

- It stores the information from where the data enters into the system and exists out of the system. The Data Store may include -
 - **Files**
 - Internal to software.
 - External to software but on the same machine.
 - External to software and system, located on different machine.

Eg.

ID	D1
Name	Customer Master
Alias	Client Master
Description	Contains a record for each customer
File Type	Computer
File Format	Database
Record Size	200
Maximum Records	45,000
Average Records	42,000
Percent Growth/Year	6%

Data Dictionary Contents

Data Store

- **Tables**
 - Naming convention
 - Indexing property

Data Set/Table Name	Customer
Copy Member	Custmast
Data Structure	Customer Record
Primary Key	Customer Number
Secondary Keys	Customer Name, Telephone, Zip Code
Comments	The Customer Master file records are copied to a history file and purged if the customer has not purchased an item within the past five years. A customer may be retained even if he or she has not made a purchase by requesting a catalog.

Decision Trees

- ❖ Decision trees are method for defining complex relationships by describing decisions and avoiding the problems in communication.
- ❖ A DT is a diagram that shows alternative actions and conditions within horizontal tree framework. Thus, it depicts which conditions to consider first, second, and so on.

Decision Trees

- Decision trees depict the relationship of each condition and their permissible actions.
- A square node indicates an action and a circle indicates a condition.
- It forces analysts to consider the sequence of decisions and identifies the actual decision that must be made.

Example : Library Membership Automation Software

- ❖ Library Membership Automation Software (LMS) where it ought to support the following three options: New member, Renewal, and Cancel membership.
- **New Member Option:**
 - Decision:**

Once the 'new member' possibility is chosen, the software system asks for details concerning the member just like the member's name, address, number, etc.
 - Action:**

If correct info is entered then a membership record for the member is made and a bill is written for the annual membership charge and the protection deposit collectible.

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- **Renewal Option:**

Decision:

If the 'renewal' possibility is chosen, the LMS asks for the member's name and his membership range to test whether or not he's a sound member or not.

Action:

If the membership is valid then the membership ending date is updated and therefore the annual membership bill is written, otherwise, a slip-up message is displayed.

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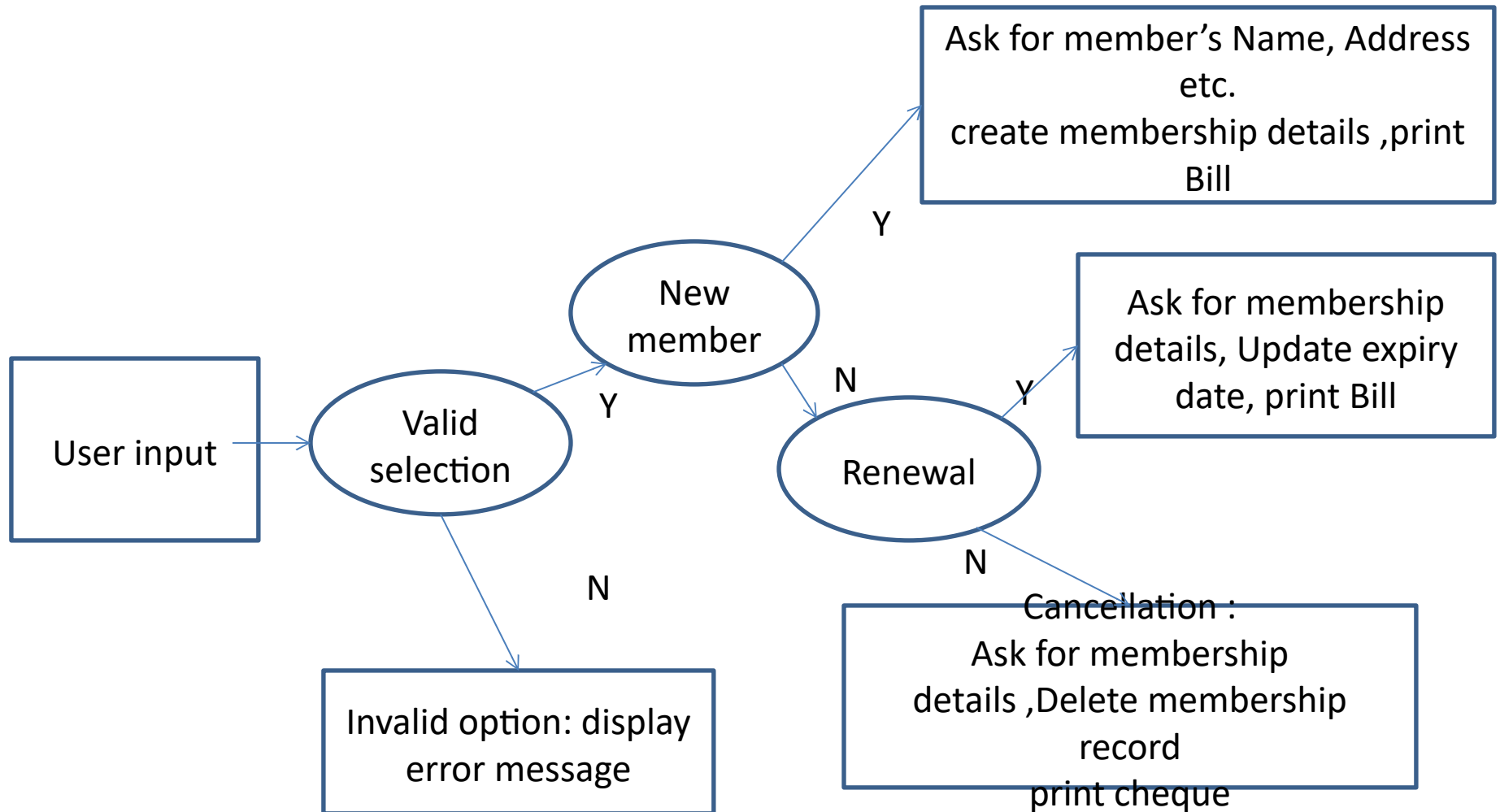
- **Cancel Membership Option:**

- **Decision:**

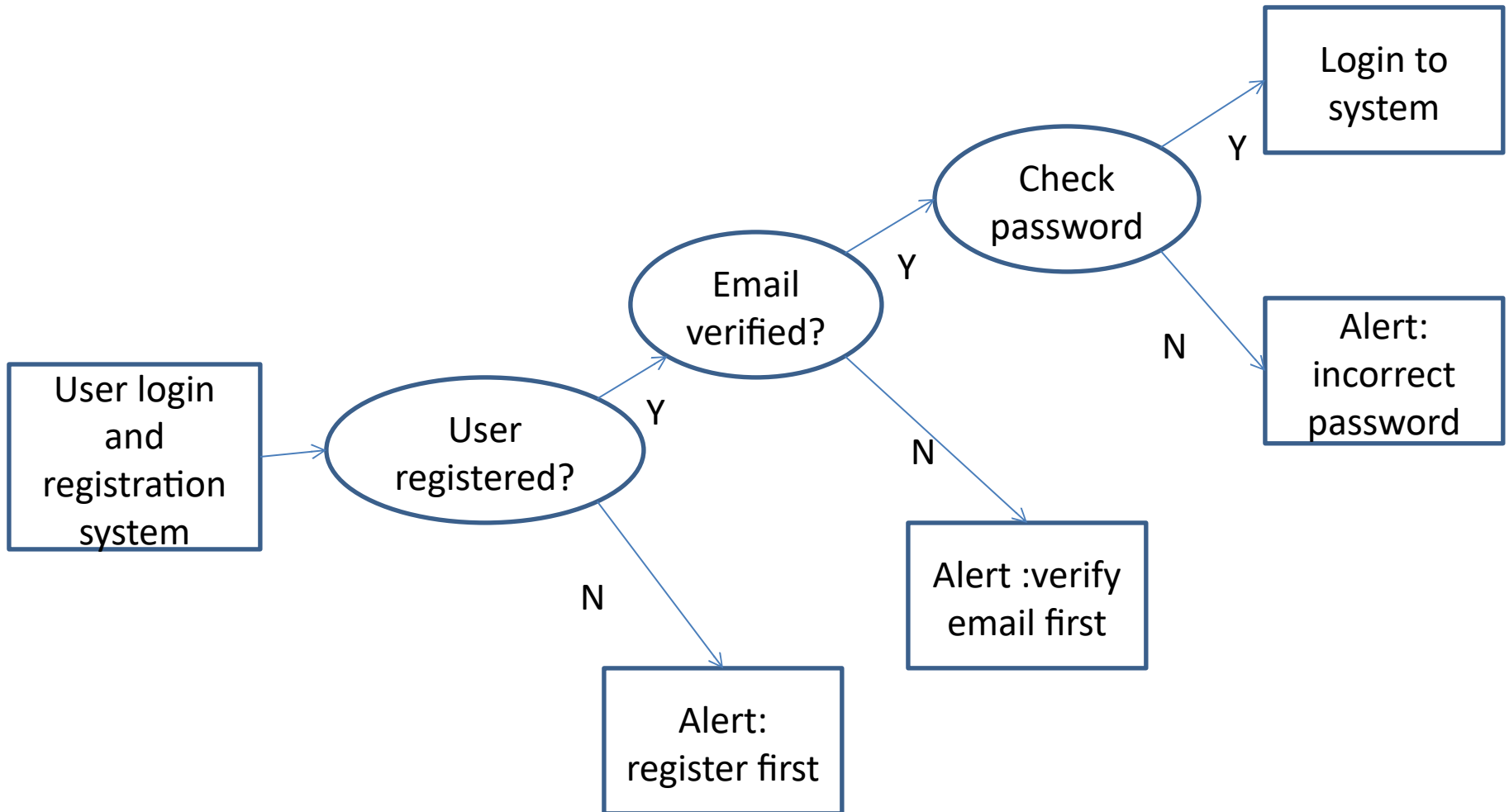
If the 'cancel membership' possibility is chosen, then the software system asks for a member's name and his membership range.

- **Action:**

The membership is off, a cheque for the balance quantity because of the member is written and at last, the membership record is deleted from the information.



- *Draw decision tree for Login and registration system!*



Decision Tables

- Decision tables are a method of describing the complex logical relationship in a precise manner which is easily understandable.
 - It is useful in situations where the resulting actions depend on the occurrence of one or several combinations of independent conditions.
 - It is a matrix containing row or columns for defining a problem and the actions.

Components of a Decision Table

- **Condition Stub** – It is in the upper left quadrant which lists all the condition to be checked.
- **Action Stub** – It is in the lower left quadrant which outlines all the action to be carried out to meet such condition.
- **Condition Entry** – It is in upper right quadrant which provides answers to questions asked in condition stub quadrant.
- **Action Entry** – It is in lower right quadrant which indicates the appropriate action resulting from the answers to the conditions in the condition entry quadrant.

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- The **entries** in decision table are given by **Decision Rules** which define the relationships between combinations of conditions and courses of action.
- In rules section,
 - ✓ Y shows the **existence of a condition**.
 - ✓ N represents **the condition, which is not satisfied**.
 - ✓ A blank - against action states **it is to be ignored**.
 - ✓ X (or a check mark will do) against action states it is **to be carried out**.

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- Eg,

CONDITIONS	Rule 1	Rule 2	Rule 3	Rule 4
Advance payment made	Y	N	N	N
Purchase amount = Rs 10,000/-	-	Y	Y	N
Regular Customer	-	Y	N	-
ACTIONS				
Give 5% discount	X	X	-	-
Give no discount	-	-	X	X

Exercise

- Assume a library system and draw a decision table for it
- The library has the following rules/Policy
 - A Teacher can take a Book for full Semester
 - A Student can take a Book for two weeks
 - A Visitor can take a Book for 1 week after deposit a security Amount
 - If Book is not available then a teacher/student can reserve a book
 - If there is a delay a student is fined 50birr
 - If there is a delay a visitor is fined 100birr
 - The visitor's security amount is refunded after returning the book
 - Delay and fine rule doesn't apply for teachers

Conditions stub	Condition Entry									
Teacher	Y	Y	Y	N	N	N	N	N	N	N
Student	N	N	N	Y	Y	Y	Y	N	N	N
Visitor	N	N	N	N	N	N	N	Y	Y	Y
Issue book	Y	N	N	Y	Y	N	N	Y	N	N
Available	Y	N	-	Y	N	-	-	Y	-	-
Return	N	N	Y	N	N	Y	Y	N	Y	Y
Delay	-	-	-	-	-	N	Y	-	Y	N
Action stub	Action Entry									
1 week								X		
2 Week				X						
Full Semester	X									
Reserve a book		X			X					
Pay 50birr refine							X			
Pay 100birr refine									X	
Depose security								X		
Refund security									X	X
Return book			X			X	X		X	X

- Possible solution for the previous scenario